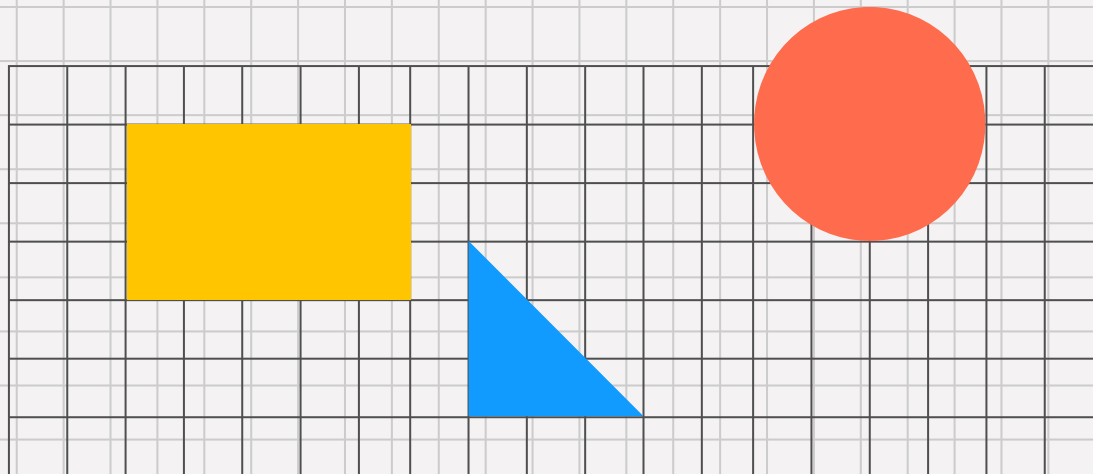
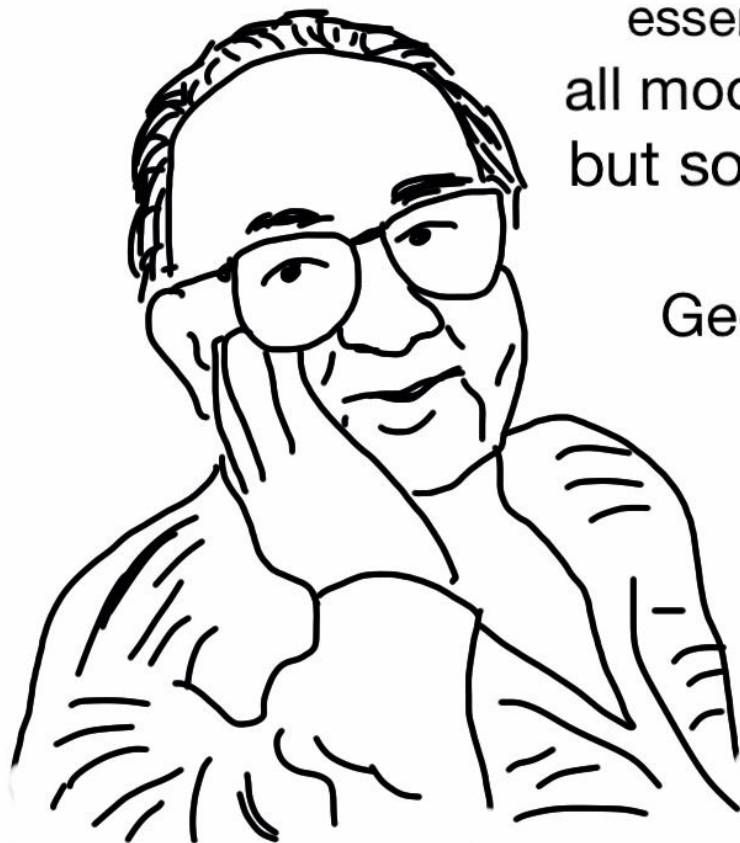


Spring 2026

IS 365

Lecture 3: Process Modeling





essentially,
all models are wrong,
but some are useful

George E. P. Box

IS365: PROJECT TEAMS

Team 1

- 1 Zuniga,Lance Nino
- 1 Mazeedian,Arman John
- 1 Lambert,Jasmine Rae
- 1 Navarro,Brogan Lucas

Team 2

- 2 Bacon,Angelina
- 2 Llamas-Cruz,Alex
- 2 Radford,Sam Jade
- 2 Tayam,Jelly Ann Manliguez

Team 3

- 3 Rogers,Ryan Erald
- 3 Sferrazza,Giovanni
- 3 Cabanero,Thomas D
- 3 Jones,Ryan

Team 4

- 4 Boulom,David David
- 4 Chew,Mason
- 4 Otey,Peyton Aric
- 4 Perez,Jaime

Team 5

- 5 Henriquez,Benji Orlando
- 5 Encinas,Kade Alexander
- 5 Carrera-Munoz,Dalia Belen
- 5 Ingle Awong,Kate Leinaala

Team 6

- 6 Nguyen,Thanh Thien
- 6 Samayoa,Liz
- 6 Thai,Annie
- 6 Kim,Mabel Grace
- 6 Victoria,Angel

Team 7

- 7 Tinajero,Tony
- 7 Li,Dylan Yen
- 7 Bayewumi,Tito David
- 7 Tang,Ryan

Team 8

- 8 Turngren,Thomas
- 8 Bassi,Anthony Celestino
- 8 Wu,Brian
- 8 Weber,Zachary Lee

Team 9

- 9 Lock,Nathan Russell
- 9 Fowler,Robert Jones
- 9 Echols,Bryce A
- 9 Previlon,Chelsie Pamela

Team 10

- 10 Trujillo Ortiz,Briana Vanesa
- 10 Duarte,Samuel Alexander
- 10 Hanson,Lauren Rebecca
- 10 Garza,Nico Luis

Team 11

- 11 Claudio,Kirra
- 11 Tordoff,Gregory
- 11 Reynolds,Jackson Romstad
- 11 Ayala,Yadira
- 11 Banga,Meha

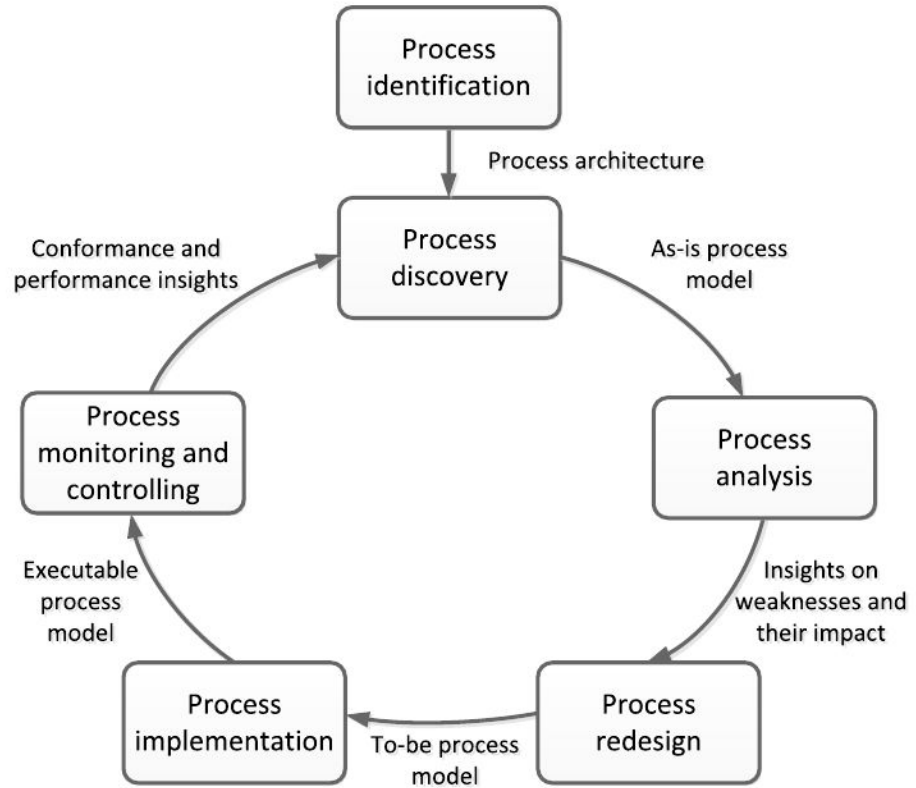
Announcements



NEWS

- 
- Assignment #1 is now posted and is due 2/9, 4 pm. Late submissions accepted until Friday for 50% credit.
 - Group Project teams posted. Pls review and let me know if there are any issues. They'll be considered finalized by 2/4.

BPM Lifecycle

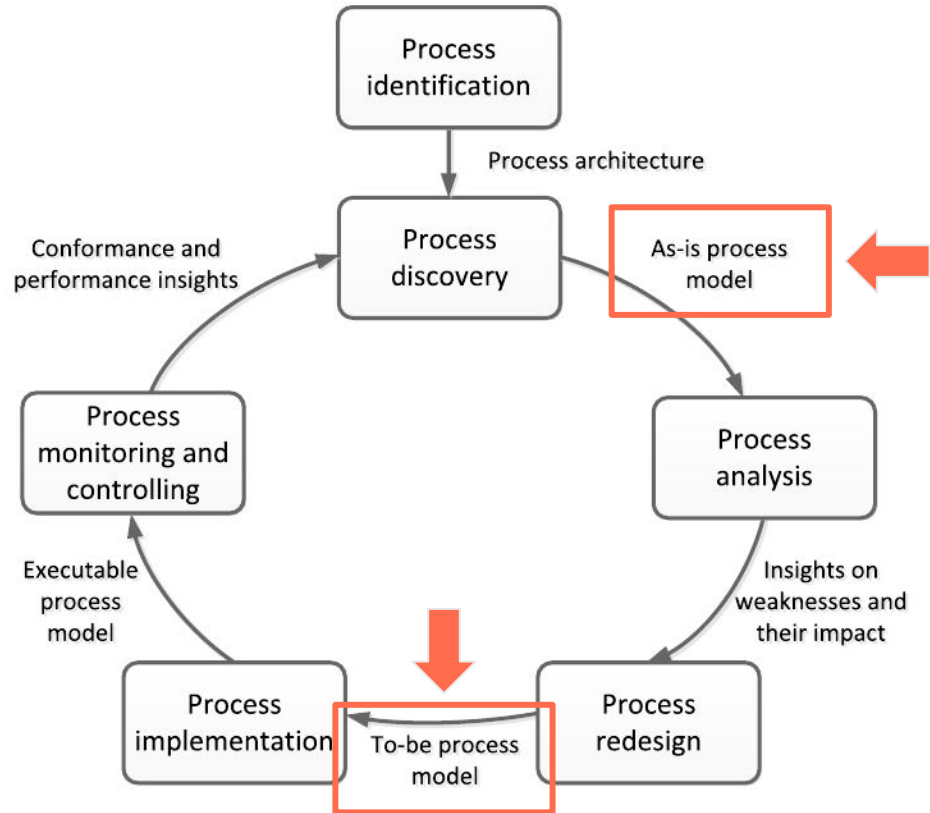


RECAP: Process Identification

- **Task:** define key processes and prioritize them
- **Output:** process architecture
- **Purpose:** to guide modeling and automation



BPM Lifecycle



Today's Topics

- BPM modeling basics
- Exclusive, parallel, and inclusive flows
- Modeling repetition with loops

Foundations of Process Modeling

Why Model Processes?

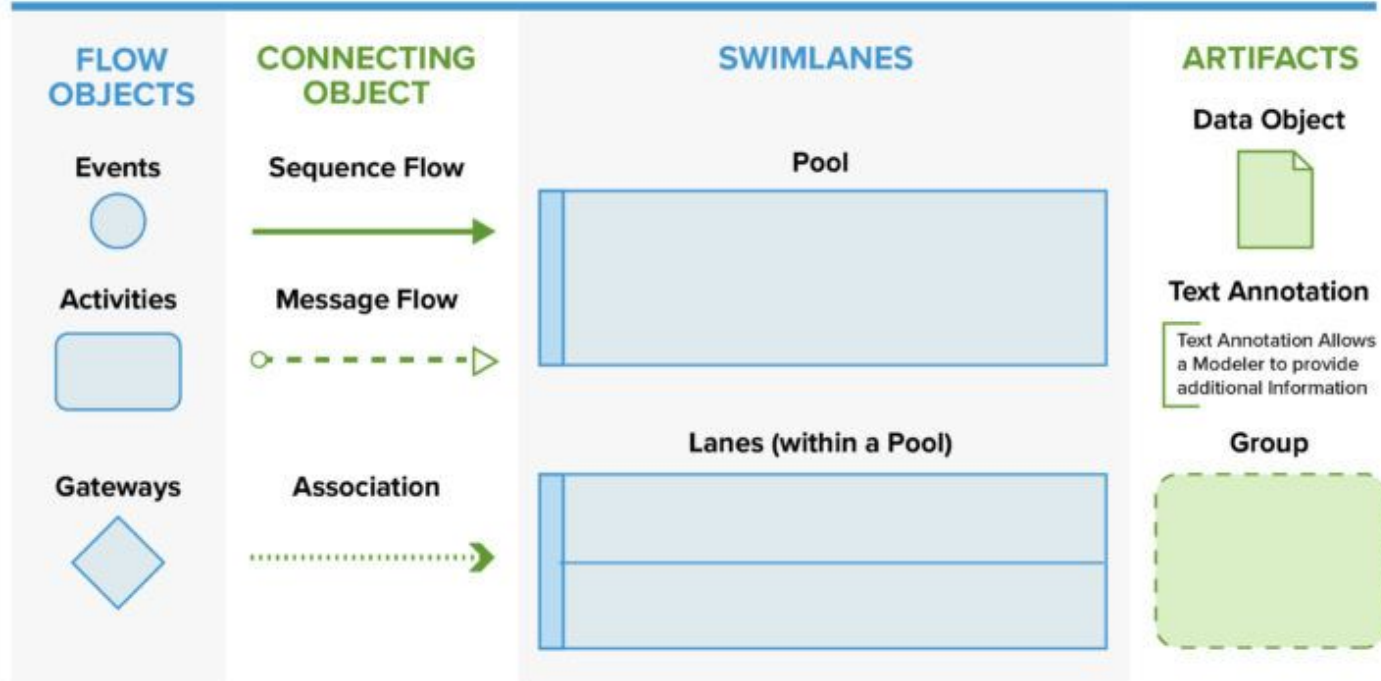
- Improve understanding of complex processes
- Communicate with stakeholders
- Identify and prevent issues
- Basis for analysis, redesign, and automation

BPMN 2.0

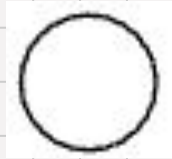
Global BPM market is \$14.4 billion. Learning BPMN can increase your value in the business world.

Business Process Model & Notation (BPMN) is a standardized diagramming language similar to Unified Modeling Language (UML) in software engineering.

Core Set of BPMN Elements



Event



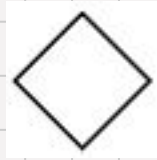
- Every BPMN diagram starts and ends with an event
- Events can also occur in the middle of a process
- Sometimes they include an icon indicating the type of event represented
- **Naming:** noun + past participle (e.g., 'Invoice emitted')
- **Remember:** Events are instantaneous

Activity



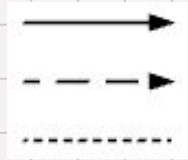
- Activities are represented by rectangles
- **Naming:** verb + noun (e.g., 'Approve order')
- **Remember:** Activities are tasks that have duration

Gateway



- Gateways split and rejoin process flows
- Sometimes they contain icons to indicate the conditions for splitting and rejoining process flows

Connector



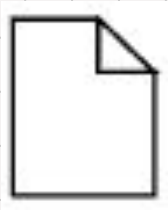
- Connectors (a.k.a. lines, processes) show the flow from one element of a BPMN diagram to the next
- Different line textures indicate the distinction between a flow of activities and a flow of messages

Pool



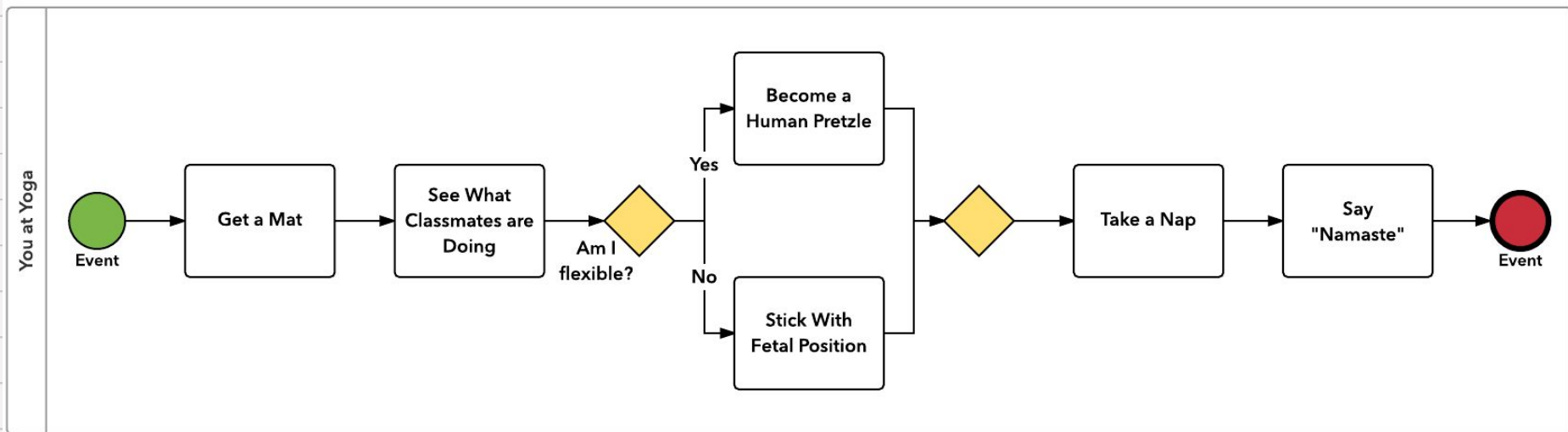
- Pools and swimlanes are used to show groups and individuals that are responsible for certain parts of a process
- Every BPMN diagram must contain at least one pool with one swim lane

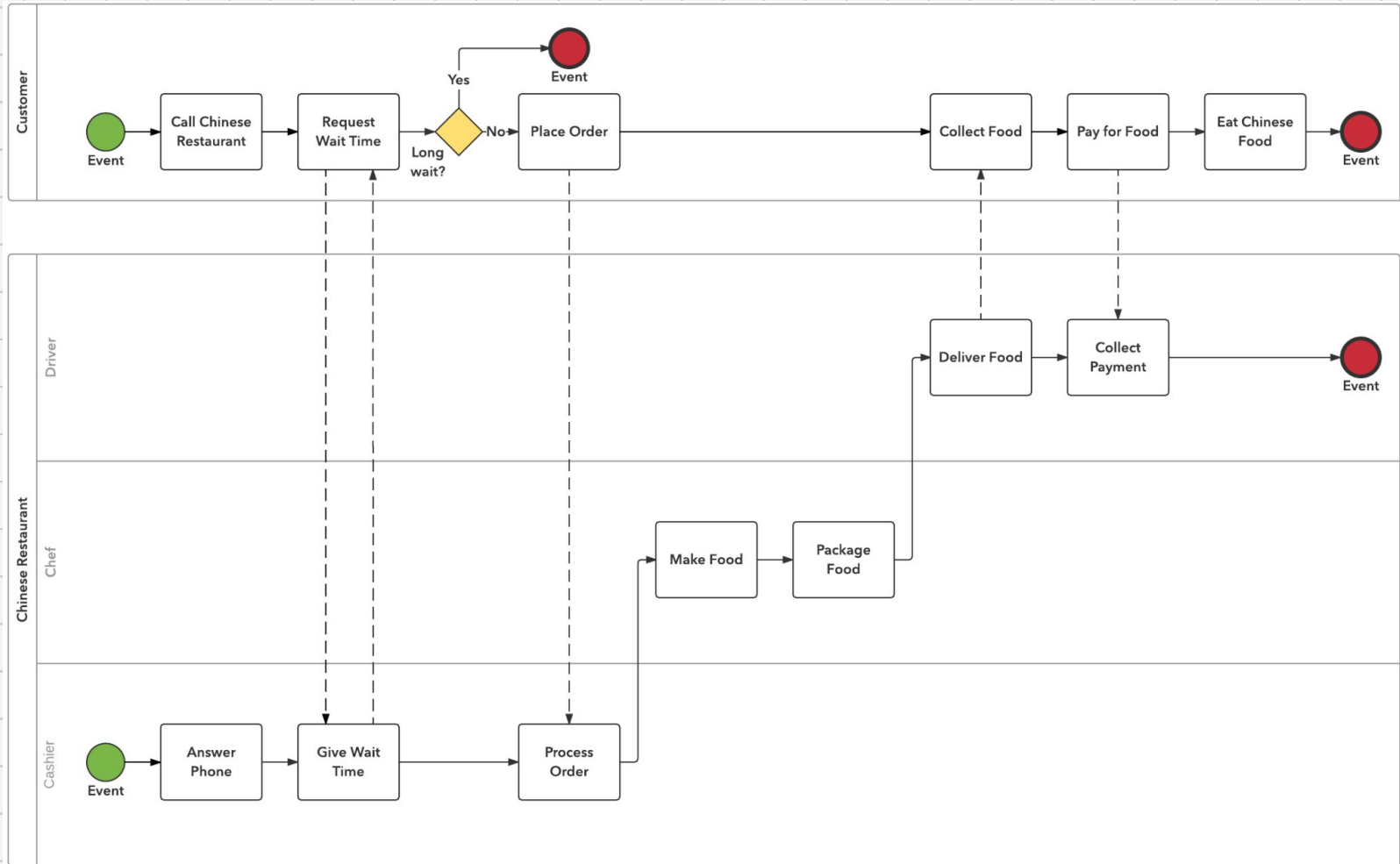
Artifact



- Artifacts add additional information to a BPMN diagram
- They include annotations, groups, and data objects

Examples / Discussion





A Few Tips

- Diagram always flows from left to right, never up and down
- Connectors should never cross over the top of each other
- Diagram the primary scenario first, then add exceptions above / below
- Include separate end events for distinct end states, e.g., successful vs. unsuccessful
- Only include one process in each pool, e.g., customer's process and restaurant's process, which interact but each have their own pool
- When two pools interact, use dashed message connectors rather than solid sequence connectors

More Tips

- Over 100 symbols exist, but you only need a core set
- Process model = blueprint (all possible executions)
- Process instance = one execution of the process
- **Tools:** Lucidchart, Visio, Google Presentation / Powerpoint

Branching and Merging

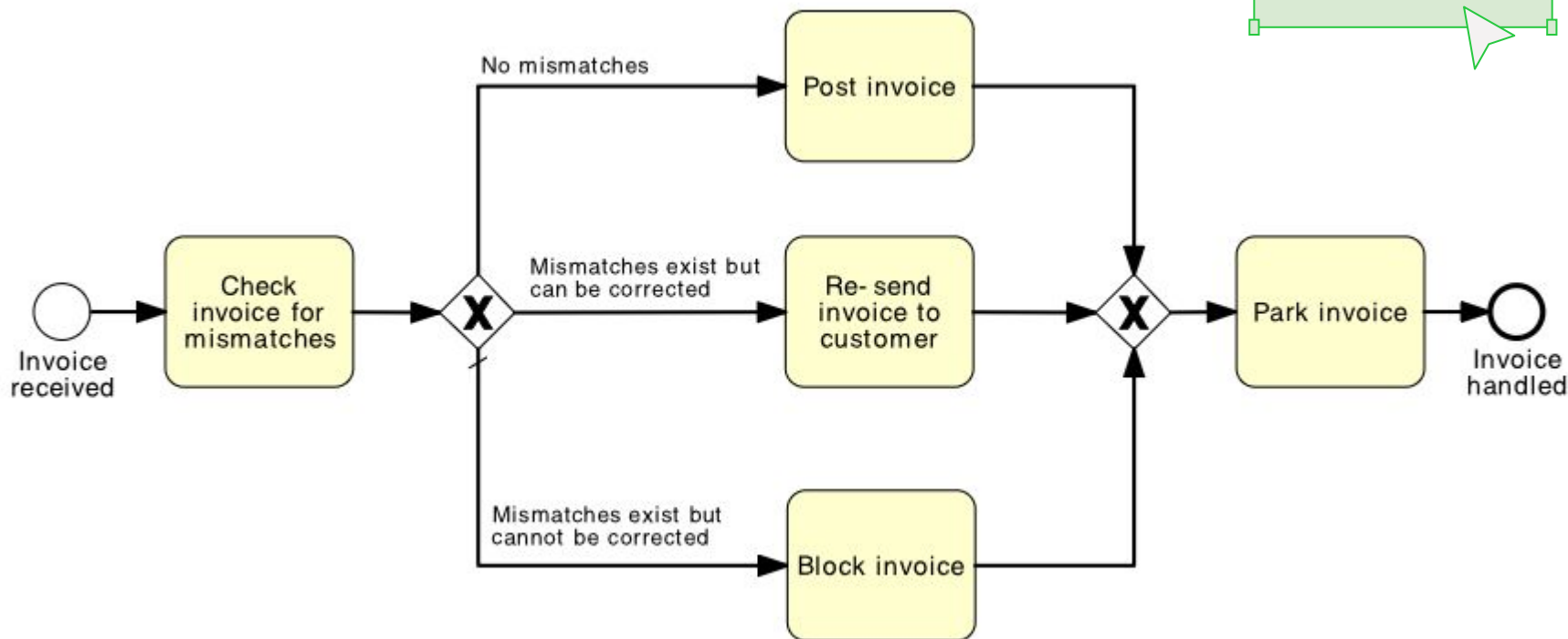
Gateways in BPMN

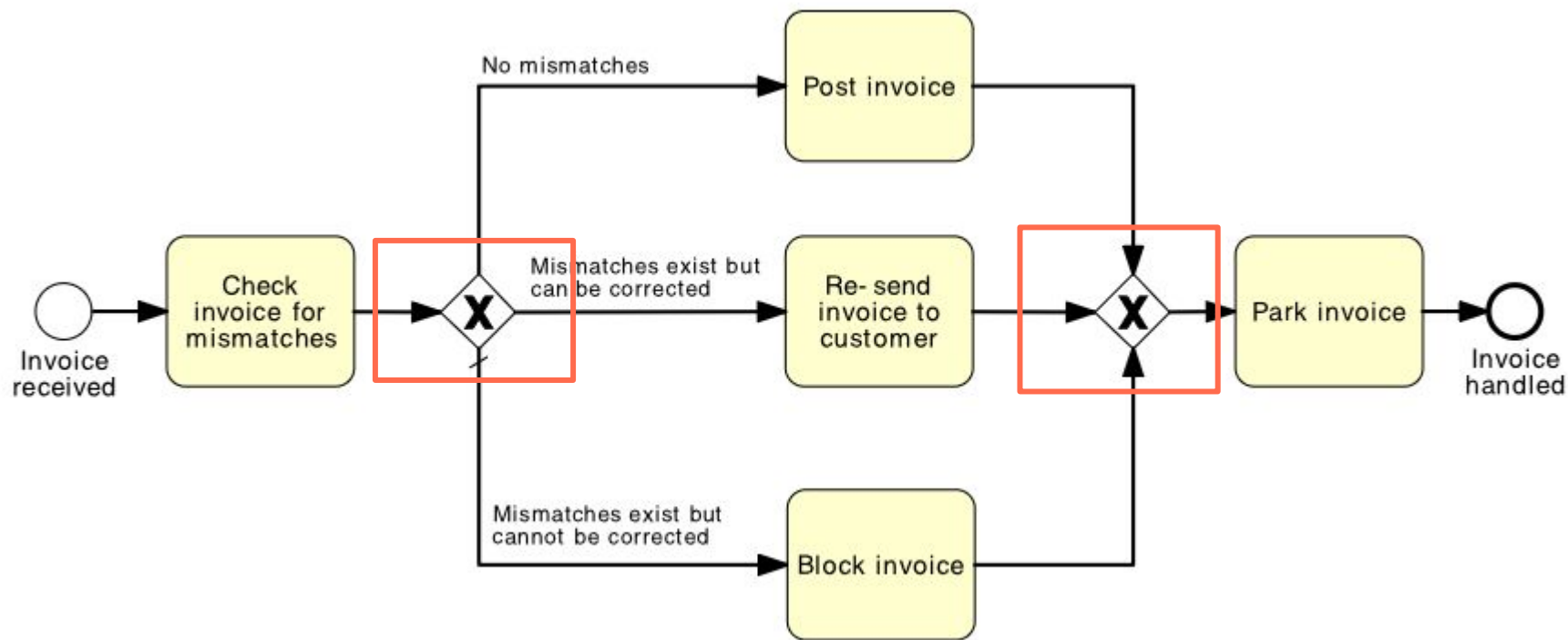
- Gateways = decision (split) or synchronization (rejoin) points
- Splits: one incoming → many outgoing flows
- Joins: many incoming → one outgoing flow

Exclusive Decisions (XOR)

- One branch taken based on conditions
- Use XOR-join to merge branches

**XOR split or
join?**

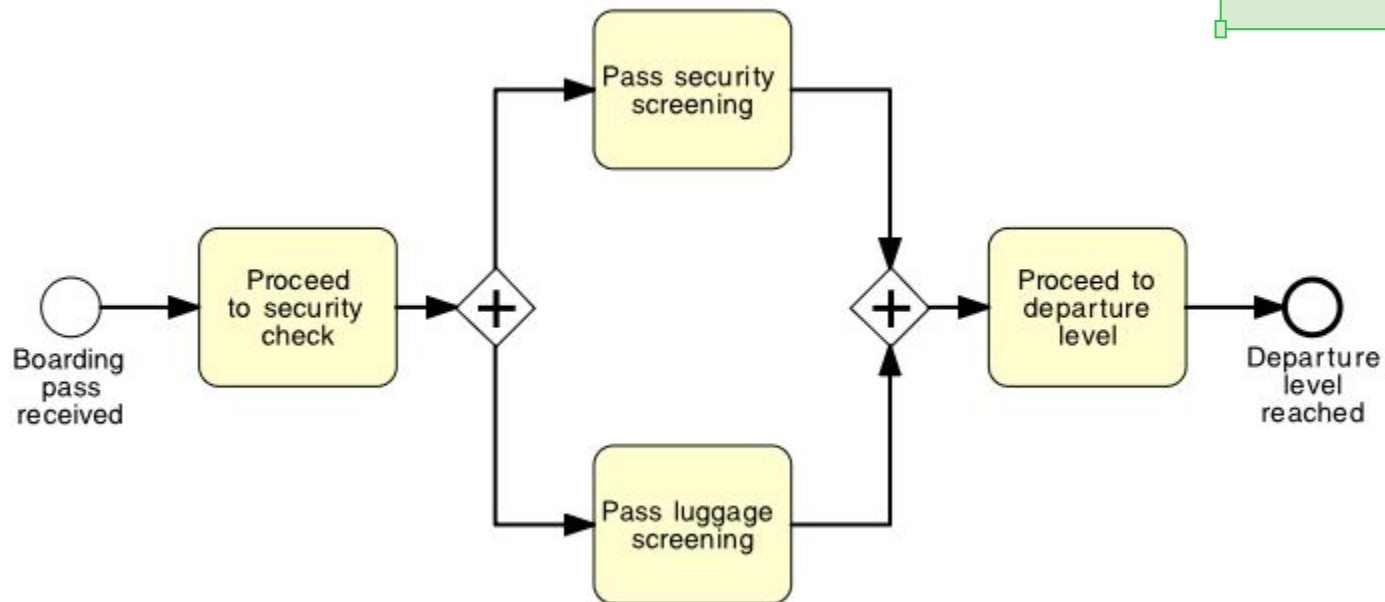


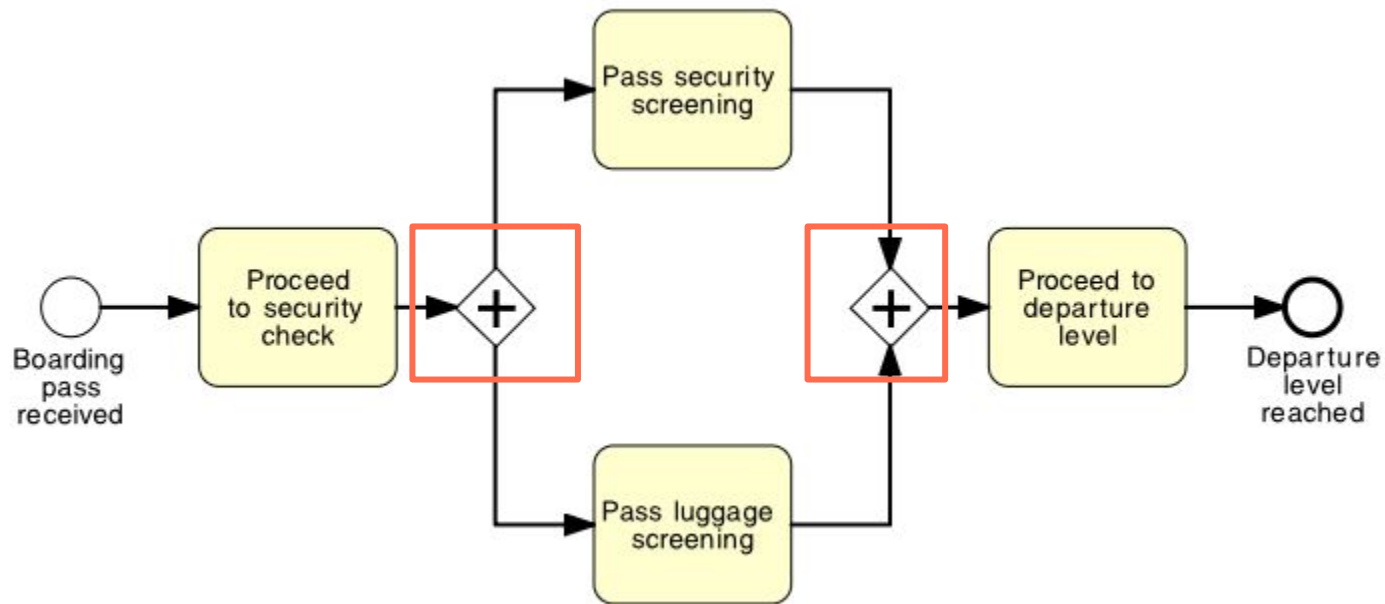


Parallel Execution (AND)

- Activities run concurrently, no dependency
- AND-split: one token becomes many
- AND-join: waits for all tokens to arrive (synchronization)

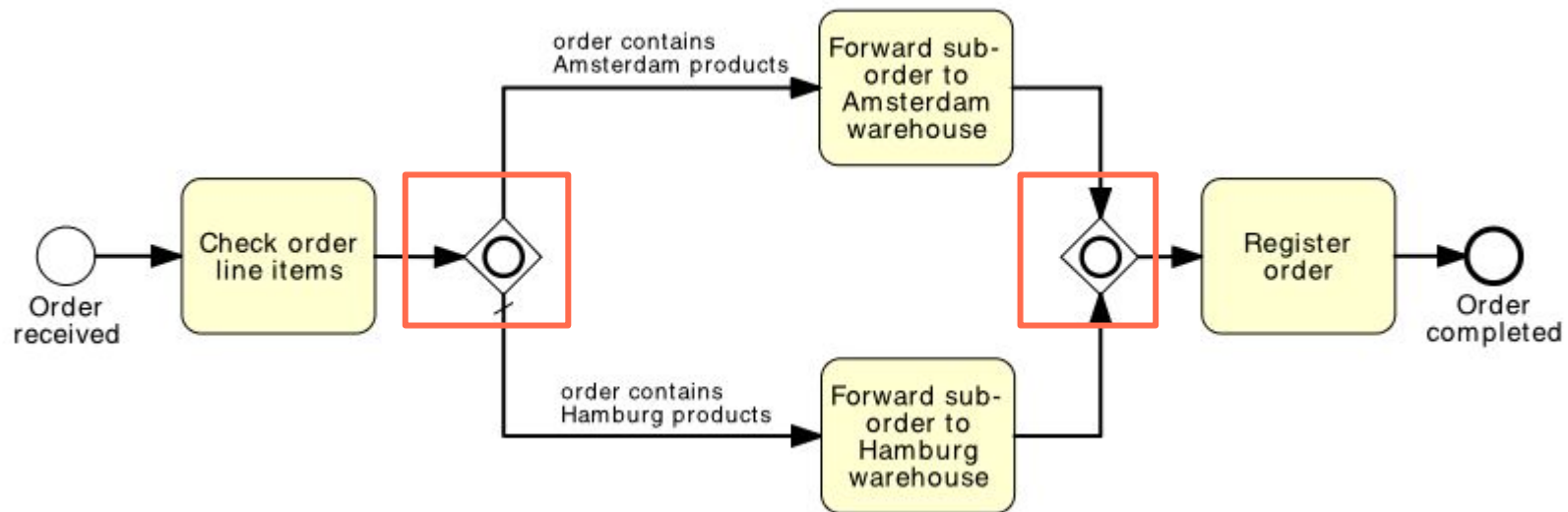
**AND split or
join?**





Inclusive Decisions (OR)

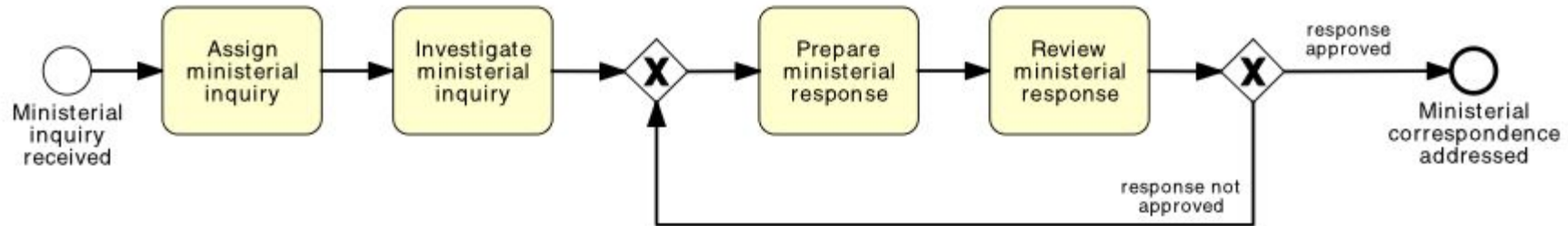
- One or more branches may be taken simultaneously
- OR-split: multiple conditions can be true
- OR-join: waits only for active branches (avoids deadlock)



Repetition

Rework and Repetition

- Looping back to repeat activities
- Requires a decision activity at the end of block
- XOR-split used for approval/rejection and repetition control



BPMN Cheat Sheet posted on Web Campus

BPMN 2.0 - Business Process Model and Notation

<http://bpmb.de/poster>

Activities

- Task**
A Task is a unit of work, the job to be performed. When marked with a symbol it indicates a Sub-Process, an activity that can be refined.
- Transaction**
A Transaction is a set of activities that logically belong together; it might follow a specified transaction protocol.
- Event Sub-Process**
An Event Sub-Process is placed into a Process or Sub-Process. It is activated when its start event gets triggered and can interrupt the higher level process context or run in parallel (non-interrupting) depending on the start event.
- Call Activity**
A Call Activity is a wrapper for a globally defined Task or Process reused in the current Process. A call to a Process is marked with a symbol.

Activity Markers

- Sub-Process Marker
- Loop Marker
- Parallel MI Marker
- Sequential MI Marker
- Ad Hoc Marker
- Compensation Marker

Task Types

- Send Task
- Receive Task
- User Task
- Manual Task
- Business Rule Task
- Service Task
- Script Task

- Sequence Flow**
defines the execution order of activities.
- Default Flow**
is the default branch to be chosen if all other conditions evaluate to false.
- Conditional Flow**
has a condition assigned that defines whether or not the flow is used.

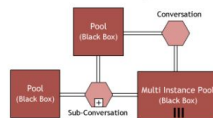
Gateways

- Exclusive Gateway**
When splitting, it routes the sequence flow to exactly one of the outgoing branches. When merging, it awaits one incoming branch to complete before triggering the outgoing flow.
- Event-based Gateway**
Is always followed by catching events or receive tasks. Sequence flow is routed to the subsequent event/task which happens first.
- Parallel Gateway**
When used to split the sequence flow, all outgoing branches are activated simultaneously. When merging parallel branches it waits for all incoming branches to complete before triggering the outgoing flow.
- Inclusive Gateway**
When splitting, one or more branches are activated. All active incoming branches must complete before merging.
- Exclusive Event-based Gateway**
Each occurrence of a subsequent event starts a new process instance.
- Complex Gateway**
Complex merging and branching behavior that is not captured by other gateways.
- Parallel Event-based Gateway**
The occurrence of all subsequent events starts a new process instance.

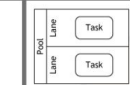
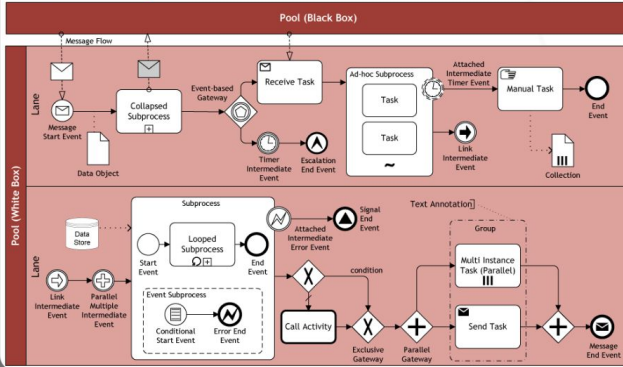
Conversations

- A Conversation defines a set of logically related message exchanges. When marked with a symbol it indicates a Sub-Conversation, a compound conversation element.
- A Call Conversation is a wrapper for a globally defined Conversation or Sub-Conversation. A call to a Sub-conversation is marked with a symbol.
- A Conversation Link connects Conversations and Participants.

Conversation Diagram



Collaboration Diagram



Pools (Participants) and Lanes represent responsibilities for activities in a process. A pool or a lane can be an organization, a role, or a system. Lanes subdivide pools or other lanes hierarchically.

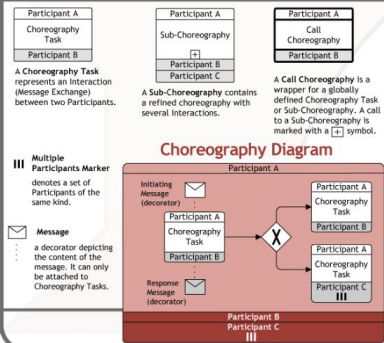
Swimlanes

Message Flow symbolizes information flow across organizational boundaries. Message flow can be attached to pools, activities, or message events. The Message flow can be decorated with an envelope depicting the content of the message.

Swimlanes

The order of message exchanges can be specified by combining message flow and sequence flow.

Choreographies



Events

	Start	Intermediate	End
Standard			
Event Sub-Process Interrupting			
Event Sub-Process Non-Interrupting			
Catching			
Boundary			
Escalating			
Timer			
Throwing			
Standard			

None: Untyped events, indicate start point, state changes or final states.

Message: Receiving and sending messages.

Timer: Cyclic timer events, points in time, time spans or timeouts.

Escalation: Escalating to an higher level of responsibility.

Conditional: Reacting to changed business conditions or integrating business rules.

Link: Off-page connectors. Two corresponding link events equal a sequence flow.

Error: Catching or throwing named errors.

Cancel: Reacting to cancelled transactions or triggering cancellation.

Compensation: Handling or triggering compensation.

Signal: Signalling across different processes. A signal thrown can be caught multiple times.

Multiple: Catching one out of a set of events. Throwing all events defined.

Parallel Multiple: Catching all out of a set of parallel events.

Terminate: Triggering the immediate termination of a process.

Data



- A **Data Object** represents information flowing through the process, such as business documents, e-mails, or letters.
- A **Collection Data Object** represents a collection of information, e.g., a list of order items.
- A **Data Input** is an external input for the entire process. A kind of input parameter.
- A **Data Output** is data result of the entire process. A kind of output parameter.
- A **Data Association** is used to associate data elements to Activities, Processes and Global Tasks.
- A **Data Store** is a place where the process can read or write data, e.g., a database or a filing cabinet. It persists beyond the lifetime of the process instance.